Homework 4

MATH 372

① $4x + 2y = 16$
$x - 5y = 70$

$\Rightarrow \frac{R_1}{2} \to R_1$
$-2R_2 \to R_2$ $\Rightarrow$ $2x + y = 8$
$-2x + 10y = -140$
$11y = -132$

$\Rightarrow y = \frac{-132}{11} \Rightarrow \boxed{y = -12}$

$4x = 16 - 2y \Rightarrow 4x = 16 + 24 \Rightarrow 4x = 40 \Rightarrow \boxed{x = 10}$

$\therefore (10, -12)$ The system is consistent and independent

② $-3x + 5y = 2$
$9x - 15y = 6$

$\Rightarrow 3R_1 \to R_1$ $-9x + 15y = 6$
$9x - 15y = 6$
$0 = 12$

$\therefore$ The system is inconsistent and it has no solution

③ $25x - 75y = 100$
$-10x + 30y = -40$

$\Rightarrow \frac{R_1}{25} \to R_1$
$\frac{R_2}{10} \to R_2$ $x - 3y = 4$
$-x + 3y = -4$
$0 = 0$

$\therefore$ The system is consistent and dependent.

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$x - 3y = 4$ let $x = t$,

$t - 3y = 4 \Rightarrow -3y = 4 - t \Rightarrow \boxed{y = \frac{t-4}{3}}$

$\therefore \left(t, \frac{t-4}{3}\right)$ are the solution.

(4)

Number of Box	$ Per box	$ for number box
S	7	7S
d	10	10d
135		1110

$S + d = 135$

$7S + 10d = 1110$

$\xrightarrow{-7R_1 \to R_1}$

$-7S - 7d = -945$

$7S + 10d = 1110$

$3d = 165$

$\boxed{d = 55}$

$S + d = 135 \Rightarrow S = 135 - 55 \Rightarrow \boxed{S = 80}$

$\therefore$ 80 boxes of standard strawberries and 55 boxes of delux strawberries are sold.

(5)

$2x + y - z = -8$

$-x + y + z = 3$

$-2x + 4z = 18$

$\Leftrightarrow$

2	1	−1	−8
−1	1	1	3
−2	0	4	18

$-R_1 \Leftrightarrow R_2$

1	−1	−1	−3
2	1	−1	−8
−2	0	4	18

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$-2R_1 + R_2 \to R_2$, $2R_1 + R_3 \to R_3$

$$\begin{bmatrix} 1 & -1 & -1 & -3 \\ 0 & 3 & 1 & -2 \\ 0 & -2 & 2 & 12 \end{bmatrix} \xrightarrow{2R_2 + 3R_3 \to R_3} \begin{bmatrix} 1 & -1 & -1 & -3 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 8 & 32 \end{bmatrix}$$

$\frac{R_3}{8} \to R_3$

$$\begin{bmatrix} 1 & -1 & -1 & -3 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix} \xrightarrow{R_2 - R_3 \to R_2} \begin{bmatrix} 1 & -1 & -1 & -3 \\ 0 & 3 & 0 & -6 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

$\frac{R_2}{3} \to R_2$

$$\begin{bmatrix} 1 & -1 & -1 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix} \xrightarrow{R_2 + R_1 \to R_1} \begin{bmatrix} 1 & 0 & -1 & -5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix} \xrightarrow{R_3 + R_1 \to R_1}$$

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

$\therefore (x, y, z) = (-1, -2, 4)$

$\therefore$ The system is consistent and independent.

⑥ $-x + 2y + 5z = 4$

$x \quad - 2z = 0$

$4x \quad -2y - 11z = 2$

$$\Leftrightarrow \begin{bmatrix} -1 & 2 & 5 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & -2 & -11 & 2 \end{bmatrix}$$

$R_1 + R_2 = R_2$, $4R_1 + R_3 \to R_3$

$$\begin{bmatrix} -1 & 2 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 6 & 9 & 18 \end{bmatrix} \xrightarrow{-R_2 + \frac{R_3}{3} \to R_3} \begin{bmatrix} -1 & 2 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

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From R_3 : $0 = 2$ so The system is inconsistent and it has no solution.

⑦

$$\begin{aligned} x - 2y + z &= 3 \\ 2x - 5y + 6z &= 7 \\ 2x - 3y - 2z &= 5 \end{aligned} \quad \Leftrightarrow \quad \left[\begin{array}{cccc} 1 & -2 & 1 & 3 \\ 2 & -5 & 6 & 7 \\ 2 & -3 & -2 & 5 \end{array}\right] \xrightarrow[-2R_1+R_3 \to R_3]{-2R_1+R_2 \to R_2} \left[\begin{array}{cccc} 1 & -2 & 1 & 3 \\ 0 & -1 & 4 & 1 \\ 0 & 1 & -4 & -1 \end{array}\right] \xrightarrow{R_2+R_3 \to R_3} \left[\begin{array}{cccc} 1 & -2 & 1 & 3 \\ 0 & -1 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

From R_3: $0 = 0$ so The system is consistend and dependent.

Let $\boxed{z = t}$ $\Rightarrow -y + 4z = 1$

$\Rightarrow \boxed{y = 4t - 1}$ Then $x - 2y + z = 3 \Rightarrow$

$\Rightarrow x - 2(4t-1) + t = 3 \Rightarrow x = 3 + 8t - 2 - t$

$\Rightarrow \boxed{x = 7t + 1}$

$$\boxed{(x, y, z) = (7t+1, 4t-1, t)}.$$

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(8) $A + B + C = 100$

$\frac{1}{2}A + \frac{4}{5}B = B$

$\frac{1}{5}B + \frac{2}{3}C = C$

$\frac{1}{3}C + \frac{1}{2}A = A$

$\frac{1}{2}A + \frac{4}{5}B = B \Rightarrow 5A + 8B = 10B \Rightarrow 2B = 5A \Rightarrow \boxed{\frac{5}{2}A = B}$

(or) $\boxed{\frac{2B}{5} = A}$

$\frac{1}{5}B + \frac{2}{3}C = C \Rightarrow 3B + 10C = 15C \Rightarrow 5C = 3B \Rightarrow \boxed{C = \frac{3}{5}B}$

(or) $\boxed{B = \frac{5}{3}C}$

$\frac{1}{3}C + \frac{1}{2}A = A \Rightarrow 2C + 3A = 6A \Rightarrow 2C = 3A \Rightarrow \boxed{A = \frac{2}{3}C}$

(or) $\boxed{\frac{3}{2}A = C}$

$\frac{2}{3}C + \frac{5}{2}A + \frac{3}{5}B = 100$, $A + B + C = 100$

$\frac{2}{3}C + \frac{5}{3}C + C = 100 \Rightarrow \frac{2C + 5C}{3} + C = 100$

$\Rightarrow 7C + 3C = 300 \Rightarrow 10C = 300 \Rightarrow \boxed{C = 30}$

$A = \frac{2}{3}C \Rightarrow A = \frac{2 \times 30}{3} \Rightarrow \boxed{A = 20}$

$B = \frac{5}{3}C \Rightarrow B = \frac{5(30)}{3} \Rightarrow \boxed{B = 50}$

∴ There are 20 students in classroom A, 50 students in classroom B and 30 students in classroom C.

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